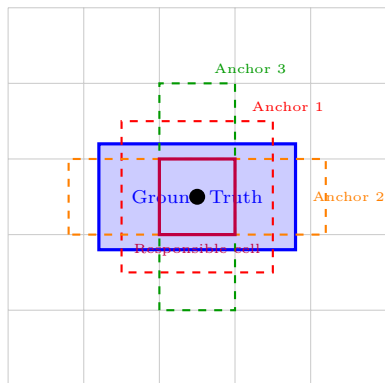


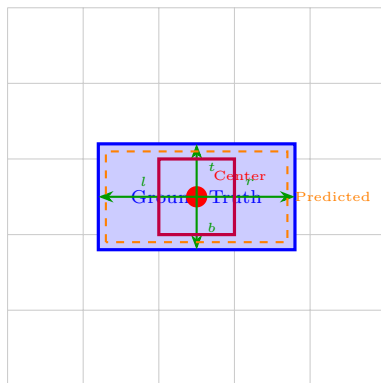
Anchor-Based vs Anchor-Free Detection

Anchor-Based (YOLOv5)



Predefined anchors with fixed aspect ratios matched to objects

Anchor-Free (YOLOv8/11)



Direct regression of distances to box edges from center point

Comparison

Aspect	Anchor-Based	Anchor-Free
Hyperparams	Many (anchor sizes)	Fewer
Novel Objects	May need tuning	Better generalization
Prediction	Offset from anchor	Direct regression